

A ladder to the septic

Hilbert's thirteenth problem through A_7 : what survives, what remains, and where new language may enter

Verified local gain. For every faithful complex A_7 -curve C with quotient signature $(2, 4, 7)$, and every involution $\tau \in A_7$,

$$\text{gon}(C/\langle \tau \rangle) \geq 10.$$

The argument is a saturated fixed-point determinant followed by a small centralizer norm. A common ramification class of degree 1296 acquires full A_7 symmetry, contradicting the free action of an order-five subgroup. It settles the involution-factor subgoal, not the septic.

1 The actual target and the rules of the game

Let $G = A_7$ permute the algebraically independent variables $x_1, \dots, x_7$, and set

$$L = \mathbb{C}(x_1, \dots, x_7), \quad K = L^G.$$

The Galois extension L/K is a generic, versal A_7 torsor. For a finite extension M/K , $\text{RD}_{\mathbb{C}}(M/K) \leq d$ means there is a finite tower $K = F_0 \subset F_1 \subset \dots \subset F_s$ with $M \subset F_s$ such that, at each step, $F_i = F_{i-1}E'_i$ for some finite extension E'_i/E_i and some $E_i \subset F_{i-1}$ with $\text{trdeg}_{\mathbb{C}} E_i \leq d$; the equality is taken in a common algebraic closure. This permits *arbitrarily many steps of arbitrary finite degrees*.

Versality gives $\text{RD}_{\mathbb{C}}(G) = \text{RD}_{\mathbb{C}}(L/K)$. The composition-factor theorem gives $\text{RD}_{\mathbb{C}}(S_7) = \text{RD}_{\mathbb{C}}(A_7)$. The established upper bound is three. Thus a complete algebraic QED for this form of Hilbert 13 is

$$\text{RD}_{\mathbb{C}}(L/K) \geq 3, \quad \text{equivalently } \text{RD}_{\mathbb{C}}(S_7) = 3.$$

Even $\text{RD}_{\mathbb{C}}(A_7) > 1$ is presently open in the literature. A *one-shot accessory* of degree n is instead one finite cover of degree at most n before compression to a faithful G -curve; $\text{ed}_{\mathbb{C}}(G; \leq n) > 1$ excludes all such covers. That property alone says nothing about unrestricted towers.

2 Status ledger: what the audit actually checked

Claim	Verification and limit
Involution quotient $\text{gon} \geq 10$	Proof reconstructed below from fixed-point counts, group generation, and cyclic descent. The 105-vertex incidence graph, 420 edges, 140 Klein triangles, and normalizer generation were independently enumerated.
Genus and marked covers	Fano-plane certificate rerun: 14 and 32 excluded candidate triples, all in order-168 subgroups. Of 192 marked $(2, 4, 7)$ pairs, 96 generate A_7 and 96 generate order 168; the generating pairs form four inner marked classes.
$\text{ed}_{\mathbb{C}}(A_7; \leq 17) > 1$	Transfer, genus-gap, connectedness, and prime-local arguments checked against their stated inputs. It extends the published degree-six bound as a manuscript deduction, without a priority claim.
Global gonality reduction	The claimed $\text{gon}(C) \geq 17$ in genus 136 and ≥ 19 on the six other low-genus rows has its arithmetic and quoted theorem ranges checked. Its refined Castelnuovo/hyperplane lifting chain deserves a separate line-by-line referee check before citation as a settled theorem.
Torsion/norm model	Useful as a conditional reconstruction. Its moving case has three candidate divisors; a universal two-equation elimination does not follow. The local quotient proof makes that collision calculation unnecessary for $\text{gon}(C/\tau) \geq 10$.

3 A compact proof certificate for the new obstruction

For any involution $v \in G$, let R_v be its reduced fixed-point divisor. In a $(2, 4, 7)$ cover $C \rightarrow C/G \simeq \mathbb{P}^1$, $g(C) = 136$, $|C_G(v)| = 24$, and the order-two and order-four branch orbits yield

$$\deg R_v = 24/2 + 24/4 = 18.$$

There is no order-five inertia, so every order-five element acts freely. The quotient $D = C/\langle \tau \rangle$ has genus 64. For two distinct commuting involutions τ, μ , with $\nu = \tau\mu$, the induced involution $\bar{\mu}$ on D fixes 18 points and the quotient map $\pi : C \rightarrow D$ satisfies $\pi^* R_{\bar{\mu}} = R_\mu + R_\nu$.

Determinant principle. If $f : Z \rightarrow \mathbb{P}^1$ has degree d and an involution v fixes r points, the determinant of the two pairs of sections defining f and $f \circ v$ lies in $H^0(Z, f^* \mathcal{O}(1) \otimes v^* f^* \mathcal{O}(1))$. It vanishes at the fixed points. If $r > 2d$ it vanishes identically, so $f \circ v = f$. If $r = 2d$ and it is nonzero, its complete zero divisor is R_v and

$$[f^* \mathcal{O}(1)] + [v^* f^* \mathcal{O}(1)] = [R_v] \quad \text{in } \text{Pic}^{2d}(Z).$$

For $\tau = (12)(34)$, the images on D of $z = (13)(24)$, $\sigma = (34)(57)$, and $\gamma\sigma$ with $\gamma = (567)$ are fixed-point-rich involutions generating $C_G(\tau)/\langle \tau \rangle$ of order 12. A map $D \rightarrow \mathbb{P}^1$ of degree $d \leq 8$ is invariant under them and therefore has $12 \mid d$, an impossibility.

Now suppose D has a degree-nine map. Its pullback has degree 18, is τ -invariant, and has line-bundle class Q . For each commuting $\mu \neq \tau$, the determinant on C vanishes at $R_\mu + R_{\tau\mu}$, of degree 36. It cannot vanish identically: invariance under the Klein four-group would make $4 \mid 18$. Hence

$$Q + \mu^* Q = [R_\mu + R_{\tau\mu}].$$

Sum this identity over the 24 members of $C_G(\tau)$, writing $M = \sum_h h^* Q$, and form $T = 24[R_\tau] + 2M$. For every Klein four-group $V = \{1, \tau, \mu, \tau\mu\}$,

$$T = 24[R_\tau + R_\mu + R_{\tau\mu}], \quad \deg T = 24 \cdot 3 \cdot 18 = 1296.$$

Use $V_0 = \{1, (12)(34), (13)(24), (14)(23)\}$ and $V_1 = \{1, (12)(34), (12)(56), (34)(56)\}$. Their normalizers fix the same T . They generate A_7 : the first contains A_4 on 1234 and (567), and the second contains (15)(26), carrying that A_4 to A_4 on 3456. The two alternating subgroups generate A_6 on 123456, then (567) gives A_7 .

Consequently $T \in \text{Pic}^{1296}(C)^{A_7}$. For a freely acting cyclic group of order n , an invariant class linearizes and descends to the quotient, so its degree is divisible by n . Here $5 \nmid 1296$. This excludes degree nine and proves $\text{gon}(D) \geq 10$. Hurwitz plus the three distinct branch orders also give $\text{Aut}(C) = A_7$; hence *no map $C \rightarrow \mathbb{P}^1$ of degree at most 18 has an involution factor.*

Why the one-shot barrier reaches degree seventeen

The accessory-to-gonality transfer used in the audit says: if an irreducible G -equivariant cover $W \dashrightarrow V$ of degree d maps dominantly to a faithful G -curve T , and G contains M acting on $\mathbb{P}^1$ with $|M| > d$, then $\text{gon}(T) \leq d$. To see the engine, choose a free M -orbit on $\mathbb{P}^1$ and an interpolating rational function ℓ ; the maps

$$\phi_v(u) = \sum_{g \in M} \ell(g^{-1}u) gv$$

give M -equivariant curves sweeping V . Pulling back W yields components whose degrees over $\mathbb{P}^1$ sum to d . Each has a nontrivial stabilizer. Some component must map nonconstantly to T , since constant images fixed by nontrivial subgroups lie in a finite set. Generic zero and pole fibers on that component have disjoint pushforwards to T ; their norm gives a function of degree at most d on T .

Here $M = A_5$, of order 60. An orbit of a degree- $m \leq 12$ function on a faithful A_7 -curve generates a faithful curve (a trivial orbit-field action would force $2520 \mid m$) of genus at most $(m-1)^2 \leq 121$, contradicting the verified minimum genus 136. Thus a connected accessory of degree at most 12 cannot compress. At primes 2 and 3, the generic A_7 torsor has essential dimension two (Sylow groups D_8 and C_3^2 , by Karpenko–Merkurjev). An accessory of degree prime to either prime cannot compress to a curve. Up to degree 17 only 6, 12 remain; these accessories are connected because A_7 has no subgroup of index at most six or index twelve. This proves the one-shot claim from the stated classical inputs. It supplies no bound on arbitrary resolvent towers.

4 The geometric bottleneck, without hiding its dependencies

For a smooth curve C write $W_m^1(C) = \{A \in \text{Pic}^m(C) : h^0(C, A) \geq 2\}$. A basepoint-free member gives a map to $\mathbb{P}^1$ of degree at most m ; a basepointed member contains a smaller pencil. Every faithful A_7 -curve has genus at least 136. Its low-genus signatures and the retained manuscript gonality reductions leave only signature $(2, 4, 7)$, $g = 136$, and degrees 17, 18.

The mechanism behind that reduction is concrete. An orbit of a degree- $m \leq 18$ function generates the full field of C and gives $g(C) \leq (m-1)^2$. For distinct degree- m functions sharing a maximal proper field of index $e \mid m$, Castelnuovo–Severi with one function outside that field gives

$$g(C) \leq \min \left\{ \frac{m^2}{e} + em - 3m + 1, \quad 2e(m/e - 1)^2 + (e - 1)^2 \right\}.$$

All $m \leq 17$ pairs on the relevant curves are birational. For $m = 18$, only the genus-136, index-two possibility survives the numerical bound; the quotient theorem above eliminates it. For three pairwise relevant degree- m pencils, independent triple products give a degree- $3m$ curve in $\mathbb{P}^7$, whereas a trilinear dependence yields two base centers of multiplicities r, s with $r + s = m$ and singularity cost $\binom{r}{2} + \binom{s}{2}$. A surface of degree at most seven cannot contain the independent equal-pencil Segre image. These facts and the valid ranges of Petrakiev’s hyperplane-section theorem produce the retained reduction. They do not rule out independent triples at $m = 17, 18$ and $g = 136$.

A branch now closed. Suppose a birational $(17, 17)$ model on $\mathbb{P}^1 \times \mathbb{P}^1$ has an ordinary point of multiplicity 16. Its arithmetic genus is 256 and this point costs $\binom{16}{2} = 120$, exactly the defect to genus 136. Blowing up the point, the linear system $(1, 1) - E$ maps birationally to $\mathbb{P}^2$; the curve maps to degree $34 - 16 = 18$. Its plane arithmetic genus is 136, so the resulting plane curve is smooth. But a smooth plane curve of degree 18 has automorphism group of order at most $6 \cdot 18^2 = 1944$ (Harui), less than $|A_7| = 2520$. Thus this ordinary-multiplicity branch is impossible. Other singular or infinitely near configurations remain open.

5 The ladder: exact statements and logical payoffs

Rung	Statement to prove	What a proof would buy
G17	For every smooth connected C with faithful A_7 -action and quotient $\mathbb{P}^1$ branched exactly at orders $(2, 4, 7)$, $W_{17}^1(C) = \emptyset$.	Excludes degree 17 and all smaller pencils on the exceptional curves.
G18	For every such C , $W_{18}^1(C) = \emptyset$. After G17, every candidate is a genuine basepoint-free degree-18 pencil.	With the other genus rows excluded, implies the degree-18 accessory barrier.
A18	For $V = \mathbb{C}^7$ with coordinate-permuting A_7 , no irreducible degree-18 A_7 -equivariant generically finite cover $W \dashrightarrow V$ admits a dominant A_7 -map to a faithful curve.	Directly yields $\text{ed}_{\mathbb{C}}(A_7; \leq 18) > 1$ using the verified lower degrees; it can bypass G17/G18.
T1	For $K = L^{A_7}$ above, exclude <i>every</i> tower in the definition of $\text{RD}_{\mathbb{C}}(L/K) \leq 1$.	First nontrivial resolvent lower bound, $\text{RD}_{\mathbb{C}}(A_7) \geq 2$.
T2	Exclude <i>every</i> such tower whose step fields E_i have transcendence degree at most two.	Together with the known upper bound, $\text{RD}_{\mathbb{C}}(S_7) = 3$: algebraic Hilbert 13.

Logic. G17 and G18 are a sufficient route to A18; A18 is also approachable directly. *A18 does not imply T1*: arbitrary one-variable towers have unbounded length and total degree. T1 does not imply T2. A direct proof of T2 may bypass every geometric rung.

6 Open interfaces for invention

The rungs specify conclusions, not preferred tools. Three mathematical interfaces currently expose the sharpest information:

1. **Incidence and residual transport.** For an involution class with r fixed points, $c = |C_G(\tau)|$, and a conjugate family of τ -invariant degree- m maps, the nonzero determinant gives

$$Q_\tau + \mu^* Q_\tau = [R_\mu + R_{\tau\mu} + E_{\tau,\mu}], \quad E_{\tau,\mu} \geq 0, \quad \deg E_{\tau,\mu} = 2(m - r).$$

Centralizer sums satisfy $T_\tau - T_\mu = [\mathcal{E}_{\tau,\mu}] - [\mathcal{E}_{\mu,\tau}] \in \text{Pic}^0(C)$ along Klein edges. At $m = r$, residuals vanish; beyond it, no balancing theorem is known. A usable extension must prove a definite obstruction or supply an explicit counterexample, with its hypotheses on incidence, noninvariance and effectivity stated. A graph-cycle identity by itself is only necessary.

2. **Entire orbits of pencils.** A minimum degree-17 pencil on a genus-136 curve has an A_7 -orbit of at least 15 classes; at degree 18 the orbit has at least 10. Two distinct classes yield a birational pair. A method may use simultaneous syzygies, Jacobian geometry, degeneration, explicit Belyi models, or a new invariant of the orbit. It must control *every* marked $(2, 4, 7)$ cover and turn the orbit constraint into G17 or G18, or construct a counterexample to those proposed rungs.
3. **A tower-stable obstruction.** To address T1 or T2, define a feature of the generic A_7 torsor with an explicit transformation law under $F_i = F_{i-1}E'_i$, $\text{trdeg}_{\mathbb{C}} E_i \leq d$. Prove it survives *all finite chains* yet vanishes if $L \subset F_s$, or derive an equivalent categorical/valuation/cohomological obstruction. This is an invitation to invent a language, not a claim that the current Picard construction already survives a tower.

7 Routes retired or bounded by evidence

- The old two-norm “moving case” has three root divisors, not two. Effectivity of one norm root alone does not produce a nine-pencil; the specified partner collision matters.
- Averaging a Picard class over all 2520 elements of A_7 introduces a factor of five and destroys the free-five degree obstruction. The 24-element centralizer average retains it.
- The involution theorem excludes degree-two factors of maps of degree at most 18; it does not exclude arbitrary degree-17 or degree-18 pencils.
- Graph connectivity is substantive. An A_5 genus-six action with signature $(2, 2, 2, 3)$ has an involution graph of five disjoint triangles; its genus-two involution quotients admit degree-three maps. A disconnected transport calculation cannot assert a single global class.
- A single small accessory obstruction cannot be extrapolated to arbitrary towers. Prime-local essential dimension also gives no resolvent lower bound: prime groups have one-variable radical towers.
- A genus bound or a numerical Brill–Noether deficit does not settle the existence of a pencil on these special covers. The singularity-budget argument has an explicit independent-triple gap at the frontier.

8 Collaboration standard and primary sources

For each future claim, record the exact quantifiers, the proof with theorem ranges, the finite certificate if any, and one attempted counterexample. Label a deduction *proved from named inputs*, *computed*, *conditional*, or *open*; keep priority separate from correctness. In particular, the manuscript global gonality chain should receive an independent referee pass before being used as a black box.

Primary inputs. Farb–Wolfson, *Resolvent degree, Hilbert’s 13th Problem and geometry*, Definitions 1.3 and 2.3, Theorem 3.3, Proposition 3.7: <https://www.math.uchicago.edu/~farb/papers/RD.pdf>. Farb–Wolfson, *Essential dimension relative to branched covers of degree at most n* , Theorem 1.6 and Lemmas 2.2, 3.1: <https://www.math.uchicago.edu/~farb/papers/essdim4new4.pdf>. Karpenko–Merkurjev, *Essential dimension of finite p -groups*: <https://www.math.ucla.edu/~merkurev/papers/esdim-final.pdf>. Petrakiev, *A step in Castelnuovo theory via Groebner bases*, Theorems 2.7, 2.17: <https://arxiv.org/pdf/math/0604517>. Harui, *Automorphism groups of smooth plane curves*, Theorems 2.1, 2.3: <https://arxiv.org/pdf/1306.5842>. Edens–Reichstein, *Hilbert’s 13th problem in prime characteristic*, introduction (complex status): <https://ems.press/journals/dm/articles/14299068>.

Provenance. The accessory and pencil programme began in Sol’s notes; Astra supplied the determinant quotient closure and ramification-transport formulation. This chart is a verification and continuation of their joint research manuscript. Names of working lemmas are provisional and assert no publication priority.